Kuwait University
College of Life Sciences

Academic Chatbot for Kuwait University

ISC 499 – Capstone Project
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Abstract

The proposed capstone project aims to present the design and development of an AI-enabled, bilingual web-based chatbot for Kuwait University. The system will aim to provide students and prospective applicants with streamlined access to verified university information, integrating official data sources into a single interactive platform. Enhanced academic planning features are offered for College of Life Sciences (CLS) students, including guidance on course selection, and prerequisite management.

The chatbot aims to have accessibility features, bilingual interaction, and transparent handling of uncertain queries, ensuring reliable and inclusive support. By automating information retrieval and centralizing knowledge, the system reduces the need for repetitive administrative inquiries, improves information accessibility, and enhances the overall student experience. This project showcases a practical application of artificial intelligence to create a scalable, sustainable, and student-centered solution.

Problem Definition

1. Problem Statement and Justification

Kuwait University students regularly require guidance on academic procedures, administrative processes, and campus-related services throughout their student life. Although this information is available through official platforms and university offices, it is often dispersed across multiple sources, making it difficult for students to quickly locate accurate and relevant answers. As a result, many students repeatedly seek assistance from academic and administrative offices across the university for common inquiries, increasing operational pressure on staff and limiting the efficiency of existing support mechanisms.

This project proposes the development of an AI-enabled chatbot designed to enhance access to university information from multiple credible sources. The chatbot will provide general guidance applicable to all Kuwait University students, while offering more detailed and structured academic information for students of the College of Life Sciences (CLS), in alignment with the project scope and available domain knowledge. By centralizing essential university-wide and CLS-specific information within a single intelligent platform, the chatbot improves accessibility,

reduces repetitive inquiries, and represents a practical and sustainable application of artificial intelligence aimed at improving efficiency and student experience within Kuwait University.

2. Project Benefits

This chatbot project offers a variety of benefits given that it addresses a real and existing problem within the academic and administrative environment of Kuwait University.

a. Benefits for Students

- Providing a single centralized platform that combines academic and administrative information related to Kuwait University.
- Enabling continuous access to information at any time, without dependence on office hours or staff availability.
- Supporting bilingual interaction in Arabic and English, allowing students to receive information in their preferred language.
- Assisting students in understanding academic procedures, including admission requirements, course registration processes, prerequisites, and campus services.
- Reducing information-seeking effort and confusion caused by having multiple websites, social media accounts, and informal communication channels.

b. Benefits for Staff and University Administration

- Reducing the load of repetitive routine inquiries directed to administrative and academic staff.
- Ensuring consistency and alignment of responses with official university policies and approved sources.
- Minimizing the spread of misinformation and contradictory guidance across departments.
- Allowing staff members to focus on complex cases that require human judgment and individualized support.

3. System Capabilities

The functionalities are structured to address common student needs while maintaining accuracy, inclusivity, and scalability. The system offers general university-wide support for both high school students aiming to pursue their further education at KU, and enrolled students at KU,

with enhanced academic details and planning support for College of Life Sciences (CLS) students, reflecting the defined scope of the project.

a. University-Wide Information Support

The chatbot provides general information applicable to all Kuwait University students, including guidance on academic procedures, administrative processes, campus facilities like registration of buses, access to KU library and university policies. This functionality enables students to quickly access essential information without relying on in-person visits or manual searches across multiple platforms. All responses are grounded in official Kuwait University sources to ensure reliability and consistency.

The chatbot will also support prospective high school students seeking admission to Kuwait University by providing general, publicly available information on admission pathways, program requirements, and minimum eligibility criteria.

b. College of Life Sciences (CLS) Academic Planning

For students at the College of Life Sciences, the chatbot delivers more detailed information in terms of academics. This includes guidance on specific courses offered on the major sheet of all departments in CLS, course prerequisites, faculty contact information, course registration procedures, and equivalent courses across different CLS departments.

The system supports academic planning by assisting students in organizing their coursework across upcoming semesters. For CLS students, the chatbot can provide guidance on course sequencing, prerequisite completion, and recommended study paths to support timely graduation. This functionality is intended as a decision-support tool and does not replace official academic advising.

c. Grade Point Average Prediction (GPA) Insight

The chatbot allows KU students to input completed or expected course grades in order to receive an estimated Grade Point Average (GPA). This feature helps students better understand the potential academic impact of their performance in a specific semester and supports informed decision-making of students regarding future planning. All GPA estimates are clearly presented as predictive and advisory, not official records.

d. Intelligent Bilingual Interaction

The system supports both Arabic and English as core languages and follows a language-mirroring policy. It automatically detects the language of the user's input and responds in the same language, ensuring a natural and inclusive interaction experience.

e. Accessibility and Inclusive Design

Accessibility is intended to be integrated in the system. The chatbot aims to include voice input, voice output, high-contrast interface modes, and screen-reader compatibility, ensuring equitable access for users with diverse needs.

f. Automated Update of Information

To reduce reliance on manual maintenance, the system is designed to support automated information updates through integration with official Kuwait University websites, verified social media accounts, and other trusted digital sources. This may be implemented using scheduled data retrieval or agent-based mechanisms, while maintaining human oversight.

g. Uncertainty Handling and Source Transparency

When the chatbot is unable to confidently answer a query, it follows a structured response strategy by requesting clarification, transparently communicating limitations, and advising the student to inquire directly from the university and/or redirecting users to official university sources. Confident responses aim to include references to official Kuwait University sources.

h. Administrative Oversight and Control

The system aims to include an administrative control panel for authorized KU staff to access. This will allow human-in-the-loop supervision as authorized administrators can manage content, review automated updates, and correct errors, ensuring long-term reliability and governance.

4. Related Work

This section reviews existing systems relevant to the proposed project. This review includes systems that are directly related to the proposed solution, such as university-specific chatbots and virtual assistants, as well as works that are indirectly related but provide valuable technical insights into areas such as conversational AI, knowledge base design, and retrieval-augmented generation.

AUK Admission chatbot

A chatbot developed to support students by providing information about programs, admission requirements, application procedures, eligibility checks, FAQs, and personalized program recommendations.

https://dspace.auk.edu.kw/server/api/core/bitstreams/f2109773-70bc-4b98-97d0-8167e99c3988/content

FalconAI: A Custom AI University Chatbot Using the ChatGPT API

Customizable AI chatbot for GUST University used to answer inquiries for students and faculty members. This chatbot Is developed using python and chat gpt API.

https://www.dpublication.com/wp-content/uploads/2024/06/04-1784.pdf

Kuwait University Research Sector Projects Unit (RSPU) Chatbot

Kuwait University has a deployed chatbot that focus on answering inquiries for the Research Sector Projects Unit (RSPU) portal. This chatbot greets users and assists with inquiries in both text and voice inputs.

https://rspu.ku.edu.kw/contact-us

AUK IT Support AI-based Virtual Assistant Chatbot (AUKbot)

AI-enabled chatbot for technical support services at AUK. This chatbot aims to address a wide array of technical queries from students, staff, and faculty, thus reducing the workload on existing IT helpdesk.

https://dspace.auk.edu.kw/server/api/core/bitstreams/9e48f412-8646-455c-9055-cdce6c68c5ed/content

Project Plan

1. Gantt Chart

Phase	Name	Start Date	End Date
Design	Project Planning and Scope Definition	02/12/2026	03/02/2026
	Requirements Gathering	13/02/2026	16/02/2026
Planning	Data Collection	15/02/2026	22/02/2026
	Dataset Preparation and Cleaning	21/02/2026	28/02/2026
	System Design and Architecture	02/03/2026	08/03/2026
Development	Frontend UI Development (Web App)	09/03/2026	15/03/2026
	Chatbot Core Logic and Retrieval System	16/03/2026	26/03/2026
	Backend & Integration Setup	26/03/2026	05/04/2026
Testing and Evaluation	Test and Adjust	05/03/2026	15/04/2026
	Final Adjustments	06/04/2026	15/04/2026

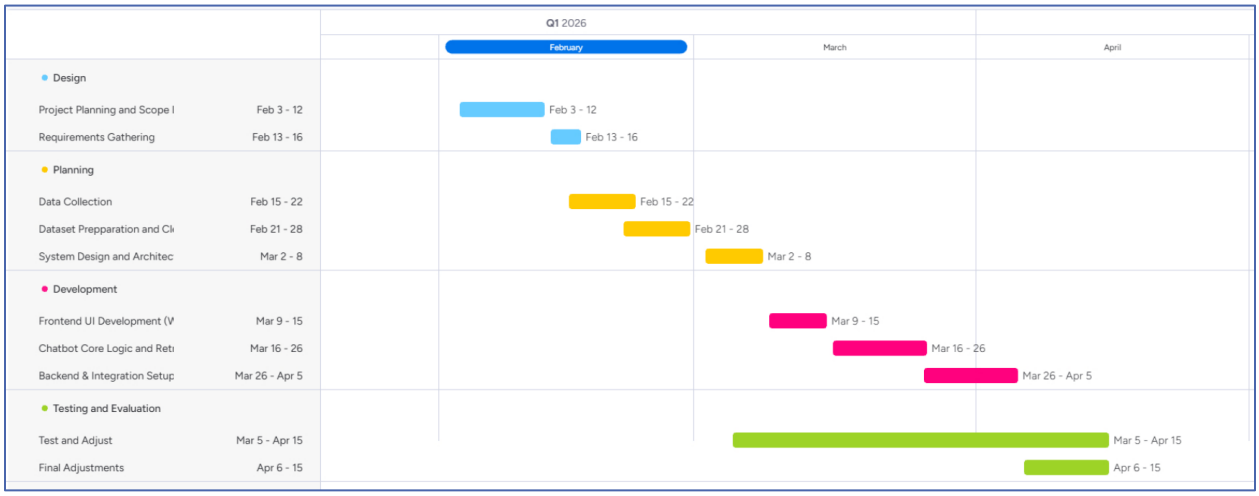

Figure 1: Gantt Chart

2. Deliverables Table

The following deliverables represent the outcomes of this capstone project:

Deliverable	Description
AI Chatbot Web Application	A fully functional AI-enabled chatbot accessible via desktop and mobile browsers, providing university-wide information for prospective and enrolled students, with enhanced academic support for College of Life Sciences (CLS) students.

Final Comprehensive Report	A detailed academic report documenting the project, including the system design, functionalities, and development process.
Final Project Presentation	A presentation summarizing the project objectives, system capabilities, and outcomes, intended to communicate the project outcomes.

3. Dependencies and Constraints

The development and deployment of the AI-enabled chatbot system rely on several technical, organizational, and user-centered dependencies. These factors are critical to the success of the project and define the boundaries within which the system can operate effectively, therefore, these factors must be acknowledged to ensure realistic expectations and boundaries of the system.

Dependencies

a. **Official University Data Sources:** The chatbot depends on access to accurate and up-to-date information from official Kuwait University sources, including the KU website, student handbooks, course catalogs, and College of Life Sciences resources.

b. **Digital Infrastructure and technical tools:** Reliable web hosting, and access to stable internet connectivity, are required for deploying the web application and supporting automated updates. Moreover, the development relies on AI frameworks, natural language processing libraries, and web development technologies compatible with both desktop and mobile environments.

c. **Administrative Oversight:** The system is partially dependent on administrative oversight. The system may be supervised by authorized KU staff to supervise content, manage data ingestion, and validate updates as part of the human-in-the-loop governance process.

d. **Student and User Engagement**: Successful use of the chatbot depends on students and prospective users actively engaging with the system and providing feedback to improve functionality.

Constraints

a. **Scope Limitation:** The system provides university-wide information to high school students seeking admission and to students enrolled in other colleges at KU. Detailed academic guidance and planning support are offered only to College of Life Sciences students.

b. **Data Availability:** The chatbot can only access information that is publicly available or provided through authorized channels, it cannot access private academic records or restricted administrative data.
c. **Data Sources Reliability:** The chatbot's responses rely on the accuracy and consistency of the information from official university sources. In cases of conflicting, incomplete, or outdated data, the system transparently communicates the uncertainty to the user and provides links or directions to the appropriate official KU resource.
d. **User Input Quality:** The chatbot's understanding depends on users entering clear, grammatically reasonable queries. Ambiguous or highly complex questions may reduce accuracy.
e. **Legal and Ethical Considerations:** The chatbot must protect user privacy by ensuring that any personal information, such as grades or other sensitive data provided by one user, is not shared with other users. The system must also avoid providing misleading information or making admission guarantees.

3. Activity Description

The project will be executed through a series of structured technical and organizational activities, aligned with the project timeline as represented in the Gantt chart.

Project Planning and Scope Definition

The project begins with defining the system scope, objectives, target users, and overall boundaries. This includes clarifying university-wide support features, enhanced academic support for College of Life Sciences (CLS) students, and system constraints to ensure realistic implementation within the semester timeframe.

Requirements Gathering

Functional and non-functional requirements will be identified and further documented. This will include chatbot capabilities, bilingual interaction, GPA estimation, academic planning guidance, accessibility considerations, and system performance expectations.

Data Collection

Relevant information will be gathered from official Kuwait University sources, including university websites, academic catalogs, student handbooks, and CLS-specific materials. Verified digital sources may also be identified to support updated information retrieval.

Dataset Preparation and Cleaning

Collected data will be reviewed, organized, structured, and cleaned to ensure consistency and accuracy. This step ensures that the chatbot retrieves reliable information and minimizes ambiguity or duplication.

System Design and Architecture

The overall system architecture will be designed, including the chatbot core logic, retrieval mechanism, backend structure, integration workflow, and administrative control considerations. The design will ensure scalability and bilingual compatibility.

Frontend UI Development (Web Application)

The web-based user interface will be developed to provide an intuitive and accessible interaction environment. The interface will be responsive and compatible with both desktop and mobile browsers.

Chatbot Core Logic and Retrieval System

The chatbot's natural language processing capabilities and information retrieval mechanisms will be implemented. This includes query interpretation, source prioritization, uncertainty handling, and structured response generation.

Backend and Integration Setup

Backend services will be configured to manage data storage, system integration, automated update mechanisms, and communication between the frontend interface and chatbot logic.

Testing and Adjustment

The system will undergo functional, usability, and performance testing to evaluate accuracy, bilingual consistency, and reliability. Identified issues will be analyzed and corrected after each phase and at the end.

Final Adjustments

Based on testing outcomes and feedback, final refinements will be made before system deployment and documentation preparation.

Application of Software Development Life Cycle (SDLC)

This project will follow an Iterative Software Development Life Cycle (SDLC) model, which allows progressive refinement through repeated cycles of planning, development, and evaluation. The iterative approach was chosen because it enables the team to build a functional prototype early, gather feedback, and make improvements in subsequent iterations. This is particularly suitable for this project, as the chatbot system requires continuous refinement of the knowledge base, response accuracy, and user interface based on testing and evaluation outcomes.

The project is structured into four iterations:

Iteration 1 (Feb 6 - Feb 12): Focuses on project planning, scope definition, and initial research. The deliverable is Report I, this deliverable, which is a documentation for the problem statement, project objectives, and project plan.

Iteration 2 (Feb 13 - Mar 1): The focus of this iteration is on gathering the system requirements, collecting data from Kuwait University sources both online and offline, and modeling the system which includes use cases and activity diagrams. The deliverable is Report II, which documents the system requirements specification.

Iteration 3 (Mar 2 - Apr 5): In this iteration we will mainly focus on system design and development, including User Interface (UI) design, database schema, architecture planning, backend development, and frontend development. The deliverables are Report III (system design specification) and Report IV (system development documentation).

Iteration 4 (Apr 5 - Apr 16): Focuses on system testing, bug fixes, final refinements. The deliverables are the Final Report.

Iteration 5 (Apr 16 – Apr 30): For this final phase we mainly focus on and preparation for demonstration and presentation. The main deliverables are project demonstration, and final presentation slides.

Conclusion

Our initial capstone report presents the proposed development of an AI-enabled university chatbot designed to centralize academic information and improve accessibility, accuracy, and student support. It outlines the problems, system features, project plan, and development approach using the SDLC framework. As this is the preliminary phase, certain technical decisions, design elements, and timelines remain tentative and may be adjusted as development progresses based on testing, implementation challenges, and feedback.